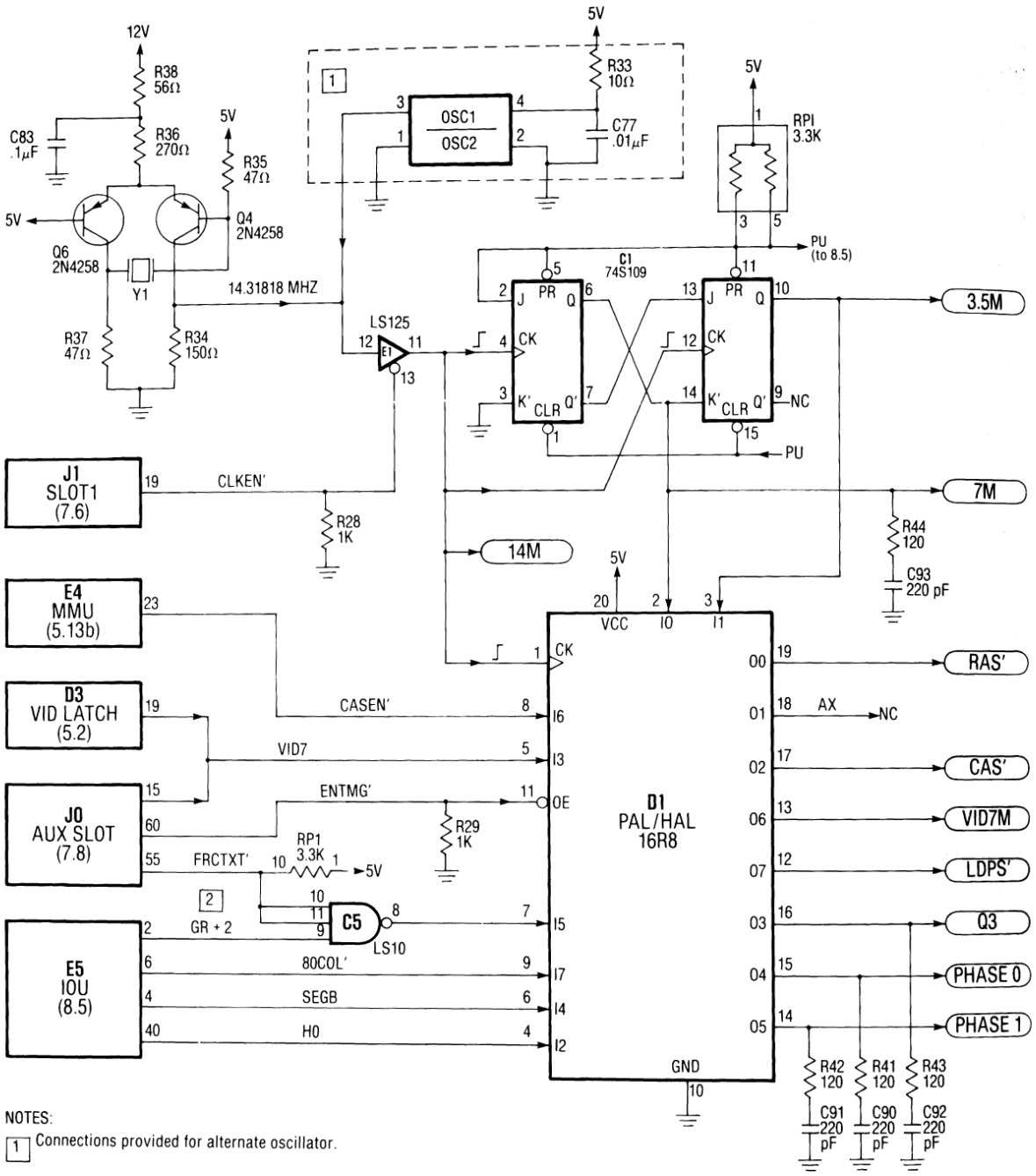
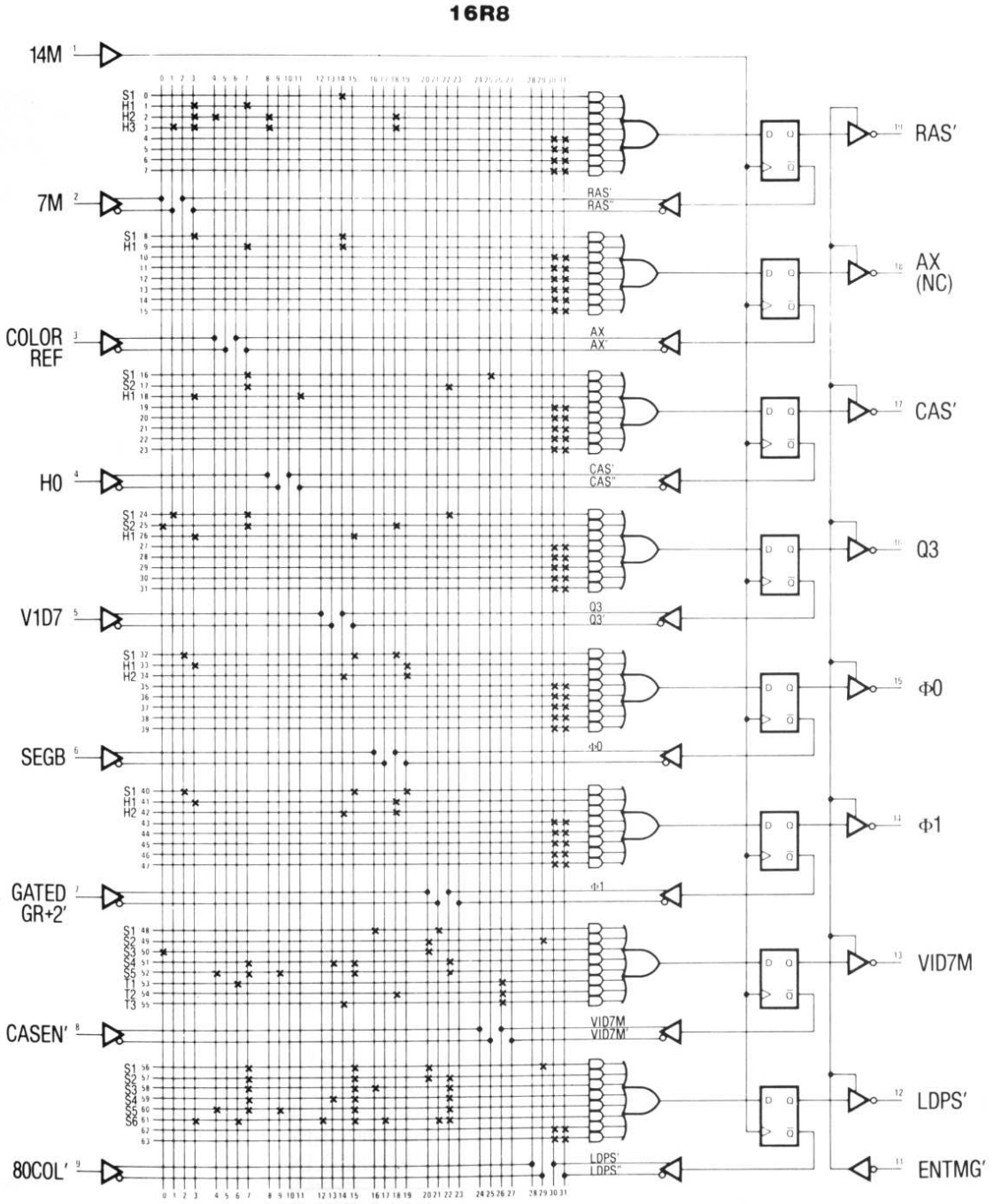


NOTES:

1 Connections provided for alternate oscillator.

2 On Revision A motherboards, E5-2 is connected directly to D1-7. The addition of FRCTXT' gating in Revision B makes the DOUBLE-RES graphics modes possible.

Figure 3.9 Schematic: The Timing Generator.



NOTE: Base drawing is from "Bipolar LSI 1984 Databook," fifth edition, reprinted with permission from Monolithic Memories, Inc. The Xs were filled in by Jim Sather.

Figure 3.10 The Timing HAL Layout.

Table 3.4 Timing HAL Logic Equations.

PIN ASSIGNMENTS

ARRAY INPUTS	OUTPUTS	OTHER INPUTS
I0-2 = 7M	Q0'-19 = RAS'	CP-1 = 14M
I1-3 = CLR REF	Q1'-18 = AX	OE'-11 = ENTMG'
I2-4 = H0	Q2'-17 = CAS'	VCC-20 = +5V
I3-5 = VID7	Q3'-16 = Q3	GND-10 = GROUND
I4-6 = SEGB	Q4'-15 = Φ0	
I5-7 = GATED GR+2'	Q5'-14 = Φ1	
I6-8 = CASEN'	Q6'-13 = VID7M	
I7-9 = 80COL'	Q7'-12 = LDPS'	

SIGNAL	EQUATIONS	NOTES
RAS'	S1 = Q3 H1= RAS'' • AX' H2= RAS'' • CLR REF • H0 • Φ0 H3= RAS'' • 7M' • H0 • Φ0	FALL AFTER Q3 RISES RISE AFTER AX RISES LONG CYCLE DELAY LONG CYCLE DELAY
AX	S1 = RAS'' • Q3 H1= AX' • Q3	FALL AFTER RAS' FALLS RISE AFTER Q3 FALLS
CAS'	S1 = AX' • CASEN'' S2 = AX' • Φ1 H1= CAS'' • RAS''	MMU, MAY I? NUTS TO MMU DURING Φ1 RISE AFTER RAS' RISES
Q3	S1 = AX' • Φ1 • 7M' S2 = AX' • Φ0 • 7M H1= Q3' • RAS''	AX' • Φ1 • CAS' ALSO WORKS CAS' NO WORKEE RISE AFTER RAS' RISES
Φ0	S1 = Φ0 • RAS' • Q3' H1= Φ0' • RAS'' H2= Φ0' • Q3	TOGGLE AT RAS' • Q3
Φ1	S1 = Φ0' • RAS' • Q3' H1= Φ0 • RAS'' H2= Φ0 • Q3	Φ0 INVERTED
VID7M	S1 = GR'' • SEGB S2 = GR' • 80COL'' S3 = GR' • 7M S4 = VID7' • Φ1 • Q3' • AX' S5 = H0' • CLR REF • Φ1 • Q3' • AX' T1 = VID7M • AX T2 = VID7M • Φ0 T3 = VID7M • Q3	LORES GRAPHICS IS HIGH SPEED DOUBLE RES IS HIGH SPEED SAME AS 7M IF NOT HIRES FRCTXT'' • 80COL' ⟶ 7M, UNDELAYED HIRES DELAY CHECK AT Φ1 • Q3' • AX' NO DELAY AT RIGHT DISPLAY EDGE TOGGLE THROUGH AX KEEP TOGGLING THROUGH Φ0 KEEP TOGGLING THROUGH Q3
LDPS'	S1 = Q3' • AX' • 80COL'' • GR' S2 = Q3' • AX' • Φ1 • GR' S3 = Q3' • AX' • Φ1 • SEGB S4 = Q3' • AX' • Φ1 • VID7' S5 = Q3' • AX' • Φ1 • CLR REF • H0' S6 = Q3' • AX • RAS'' • Φ1 • VID7 • SEGB' • GR''	DOUBLE RES CAUSES DOUBLE LDPS' TEXT MODE LORES NOT DELAYED HIRES RIGHT DISPLAY EDGE CUTOFF HIRES DELAYED LDPS'